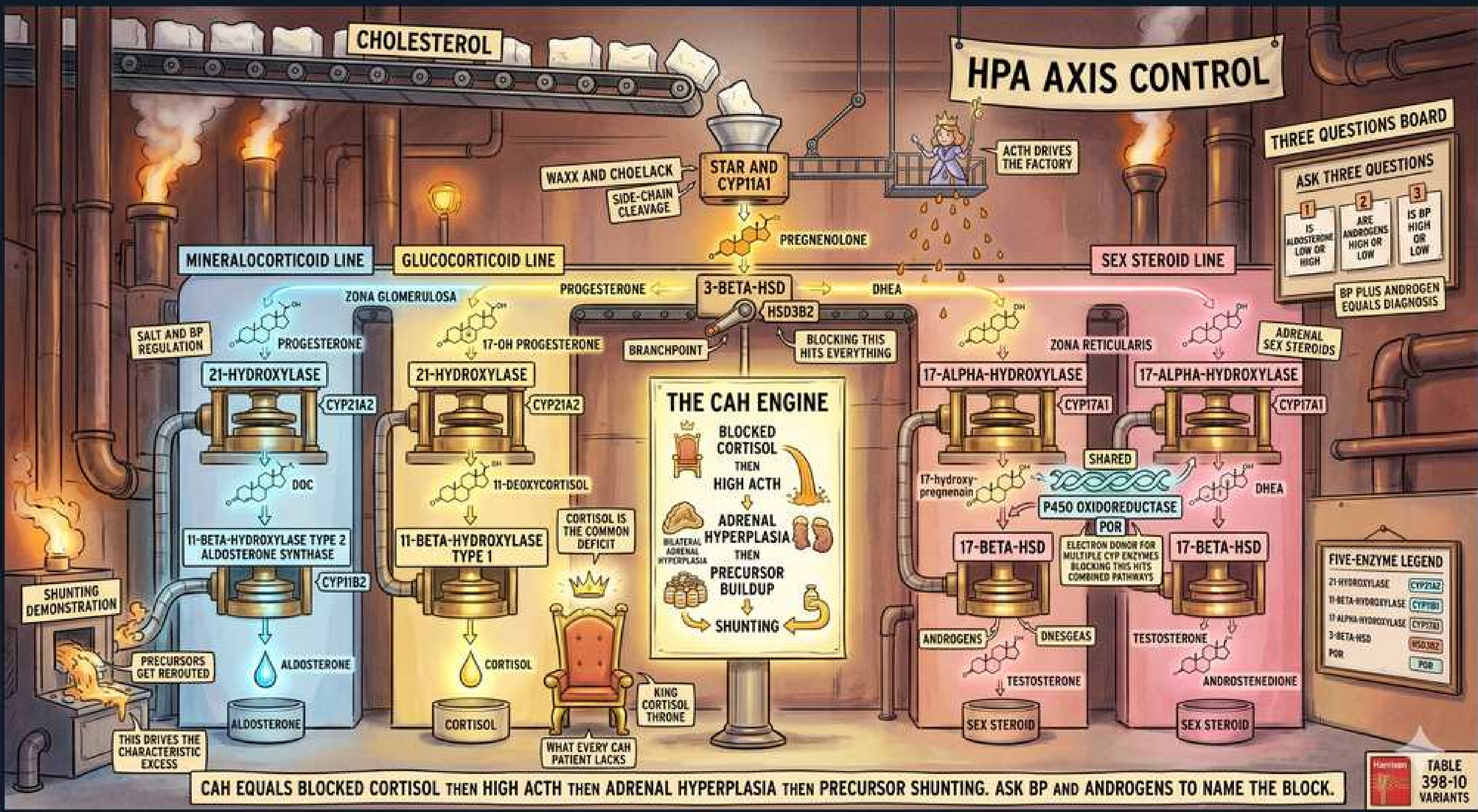
Adrenal Insufficiency/CAH — The Steroidogenesis Factory



1. Cholesterol input

WHAT YOU SEE

Cholesterol pipe feeding the whole factory from the top

WHAT IT ENCODES

Every adrenal steroid is built from cholesterol — the raw material entering the factory from the top. The committed, rate-limiting first step is StAR (steroidogenic acute regulatory protein) shuttling cholesterol into the inner mitochondrial membrane, where CYP11A1 (desmolase / side-chain cleavage) converts it to pregnenolone. Mnemonic: the whole cascade hangs off ONE input, so any upstream defect (e.g. lipoid CAH / StAR deficiency) wipes out ALL classes of steroid at once.

BOARD TRAP

ACTH does NOT act on 21-hydroxylase or any downstream enzyme directly — its trophic target is StAR/cholesterol entry. Chronic ACTH excess therefore drives hyperplasia and precursor buildup, not selective enzyme activity.

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2. StAR + CYP11A1

WHAT YOU SEE

StAR ladder and side-chain cleavage at the top, labeled ACTH drives the factory

WHAT IT ENCODES

ACTH (via the MC2R receptor and cAMP) stimulates StAR to move cholesterol into mitochondria, and CYP11A1 cleaves the side chain → pregnenolone, the parent of all three steroid classes. This is the rate-limiting committed step. StAR or CYP11A1 (lipoid CAH) deficiency = no cortisol, no aldosterone, no androgens → severe salt-wasting + under-virilized genotypic males.

BOARD TRAP

Combined loss of all steroid classes points UPSTREAM (StAR/desmolase), whereas isolated patterns localize to a single later enzyme. Don't confuse global failure with a single-enzyme CAH.

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3. Mineralocorticoid line

WHAT YOU SEE

Blue left column labeled zona glomerulosa producing aldosterone

WHAT IT ENCODES

The zona Glomerulosa is the outermost cortical layer and makes aldosterone (the 'Salt' of GFR = Salt/Sugar/Sex). It is regulated by the RAAS — angiotensin II and serum K+ — and is essentially ACTH-INDEPENDENT, which is why aldosterone is preserved in secondary/central adrenal insufficiency. Aldosterone biosynthesis requires aldosterone synthase (CYP11B2), unique to this zone.

BOARD TRAP

Aldosterone is NOT mainly ACTH-driven — it is RAAS/K+ controlled. This is exactly why central (pituitary) AI needs glucocorticoid replacement only, with no fludrocortisone.

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4. Glucocorticoid line

WHAT YOU SEE

Yellow middle column zona fasciculata making cortisol

WHAT IT ENCODES

The zona Fasciculata (middle, 'Sugar') makes cortisol and is the principal ACTH-driven zone. Path: pregnenolone → 17-OH-pregnenolone/progesterone → 11-deoxycortisol → cortisol (via 11β-hydroxylase). Cortisol exerts negative feedback on CRH/ACTH; its loss is the engine behind the high-ACTH state in both CAH and primary adrenal insufficiency.

BOARD TRAP

Loss of cortisol negative feedback raises ACTH and (in primary AI) causes hyperpigmentation via POMC/MSH — a finding ABSENT in central AI, where ACTH is low.

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5. Sex steroid line

WHAT YOU SEE

Pink right column zona reticularis making DHEA/androgens

WHAT IT ENCODES

The zona Reticularis (innermost, 'Sex') makes adrenal androgens: DHEA/DHEA-S and androstenedione, peripherally converted toward testosterone. 17α-hydroxylase then 17,20-lyase activity (both CYP17A1) divert the pathway here. Adrenal androgen excess explains the virilization/precocious puberty seen when precursors are shunted in 21-OH CAH.

BOARD TRAP

DHEA-S is essentially adrenal-specific; a markedly high DHEA-S points to an adrenal androgen source (e.g. ACC or CAH), not an ovarian one.

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6. 21-hydroxylase (CYP21A2)

WHAT YOU SEE

21-HYDROXYLASE valve appearing on both mineralo and gluco lines

WHAT IT ENCODES

21-hydroxylase (CYP21A2) deficiency is the #1 cause of CAH (>90–95% of cases). It sits on BOTH the mineralocorticoid and glucocorticoid lines, so the block lowers cortisol AND aldosterone while precursors shunt into the open androgen pathway. Classic salt-wasting form presents in neonates with hyponatremia, hyperkalemia, hypotension, and virilized female genitalia; non-classic (late-onset) presents with hirsutism/oligomenorrhea.

BOARD TRAP

21-OH deficiency does NOT raise aldosterone — it causes salt-WASTING (low Na, high K, high renin). Don't confuse it with the hypertensive CAH variants (11β / 17α).

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7. 11-beta-hydroxylase

WHAT YOU SEE

11-BETA-HYDROXYLASE valve lower on the columns, DOC labeled above it

WHAT IT ENCODES

11β-hydroxylase (CYP11B1) deficiency is the 2nd most common CAH (~5%). Cortisol and aldosterone are low, but 11-deoxycorticosterone (DOC) — a potent mineralocorticoid — accumulates upstream, causing HYPERTENSION with hypokalemia and metabolic alkalosis. Androgens still rise → virilization. Mnemonic: '11 = ele-VAT-ed BP.'

BOARD TRAP

11β-OH deficiency causes HIGH BP, the OPPOSITE of 21-OH salt-wasting. Renin is suppressed (DOC excess), not elevated.

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8. 17-OHP marker

WHAT YOU SEE

17-OH-progesterone node sitting just above the 21-hydroxylase block

WHAT IT ENCODES

17-hydroxyprogesterone (17-OHP) is the substrate immediately proximal to the 21-hydroxylase step, so it pools behind a 21-OH block. A markedly elevated 17-OHP (especially on the ACTH-stimulation test) is THE diagnostic marker of classic 21-OH-deficiency CAH and is the analyte on newborn screening.

BOARD TRAP

Elevated 17-OHP localizes specifically to 21-OH deficiency. It is NOT elevated in 11β or 17α deficiency, so use it to distinguish among the hyperandrogenic/hypertensive CAH forms.

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9. 17-alpha-hydroxylase

WHAT YOU SEE

17-ALPHA-HYDROXYLASE valve on the sex/cortisol side

WHAT IT ENCODES

17α-hydroxylase (CYP17A1) deficiency blocks cortisol AND all sex steroids, but the mineralocorticoid pathway runs unopposed → DOC/corticosterone excess → HYPERTENSION, hypokalemia, and suppressed renin. Patients have absent puberty: phenotypic females (46,XX with primary amenorrhea / 46,XY with under-virilization).

BOARD TRAP

17α deficiency causes UNDER-virilization + HTN — the only hypertensive CAH WITHOUT androgen excess. Contrast with 11β (HTN WITH virilization) and 21-OH (salt-wasting WITH virilization).

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10. The CAH engine

WHAT YOU SEE

Central CAH ENGINE box: blocked cortisol then high ACTH

WHAT IT ENCODES

Core CAH logic in one loop: enzyme block → low cortisol → loss of negative feedback → high CRH/ACTH → bilateral adrenal cortical HYPERPLASIA + massive precursor buildup → precursors shunt down whatever pathway remains open. Mnemonic chain: 'Blocked cortisol → high ACTH → hyperplasia → precursor buildup → shunting.'

BOARD TRAP

The hyperplasia and precursor accumulation are driven by the high-ACTH compensatory state — treating with glucocorticoid both replaces cortisol AND suppresses ACTH, shrinking the gland and cutting androgen overproduction.

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11. Precursor shunting

WHAT YOU SEE

Arrows showing precursors shunted/rerouted out of the engine

WHAT IT ENCODES

When an enzyme valve is shut, the backed-up substrate is rerouted into the unblocked branch. In 21-OH and 11β deficiency the open branch is androgens → virilization; in 17α deficiency androgens are blocked but mineralocorticoid precursors (DOC) accumulate → HTN. Reading the shunt direction tells you the phenotype.

BOARD TRAP

Ask 'BP + androgens' to name the block: salt-wasting + virilized = 21-OH; HTN + virilized = 11β; HTN + under-virilized = 17α.

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12. Aldosterone output

WHAT YOU SEE

Aldosterone barrel at the bottom of the blue line, salt/BP drips

WHAT IT ENCODES

Aldosterone is the mineralocorticoid end-product: it acts on the distal nephron to retain Na+/water and excrete K+/H+, defending blood pressure. It is lost in classic salt-wasting 21-OH CAH and in primary adrenal insufficiency (Addison), producing hyponatremia, hyperkalemia, and high renin.

BOARD TRAP

Aldosterone is preserved in central/secondary AI (RAAS intact) — so hyperkalemia and salt-craving point toward a PRIMARY adrenal problem, not a pituitary one.

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13. Cortisol throne

WHAT YOU SEE

Crowned cortisol throne at base of yellow column

WHAT IT ENCODES

Cortisol is the dominant glucocorticoid — it maintains gluconeogenesis, vascular tone/pressor responsiveness, and suppresses ACTH via negative feedback. Its deficiency is the unifying lesion that raises ACTH in both CAH and primary AI, and its loss causes hypoglycemia, hypotension, and the dilutional hyponatremia of cortisol-deficient ADH excess.

BOARD TRAP

In central AI the hyponatremia comes from cortisol deficiency (loss of ADH suppression), NOT from aldosterone loss — which is why K+ stays normal there.

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14. Five-enzyme legend

WHAT YOU SEE

FIVE-ENZYME LEGEND key box on the right

WHAT IT ENCODES

Five steroidogenic enzymes can be deficient (StAR/CYP11A1, 3β-HSD, 21-OH, 11β-OH, 17α-OH). The fast bedside localizer is two questions: BP (high vs low/salt-wasting) and androgens (excess vs deficient). 21-OH = low BP + high androgens; 11β = high BP + high androgens; 17α = high BP + low androgens.

BOARD TRAP

3β-HSD deficiency is the classic exam curveball: it raises DHEA (a weak androgen) yet causes salt-wasting and ambiguous genitalia in BOTH sexes — high DHEA but the boy is under-virilized.
